

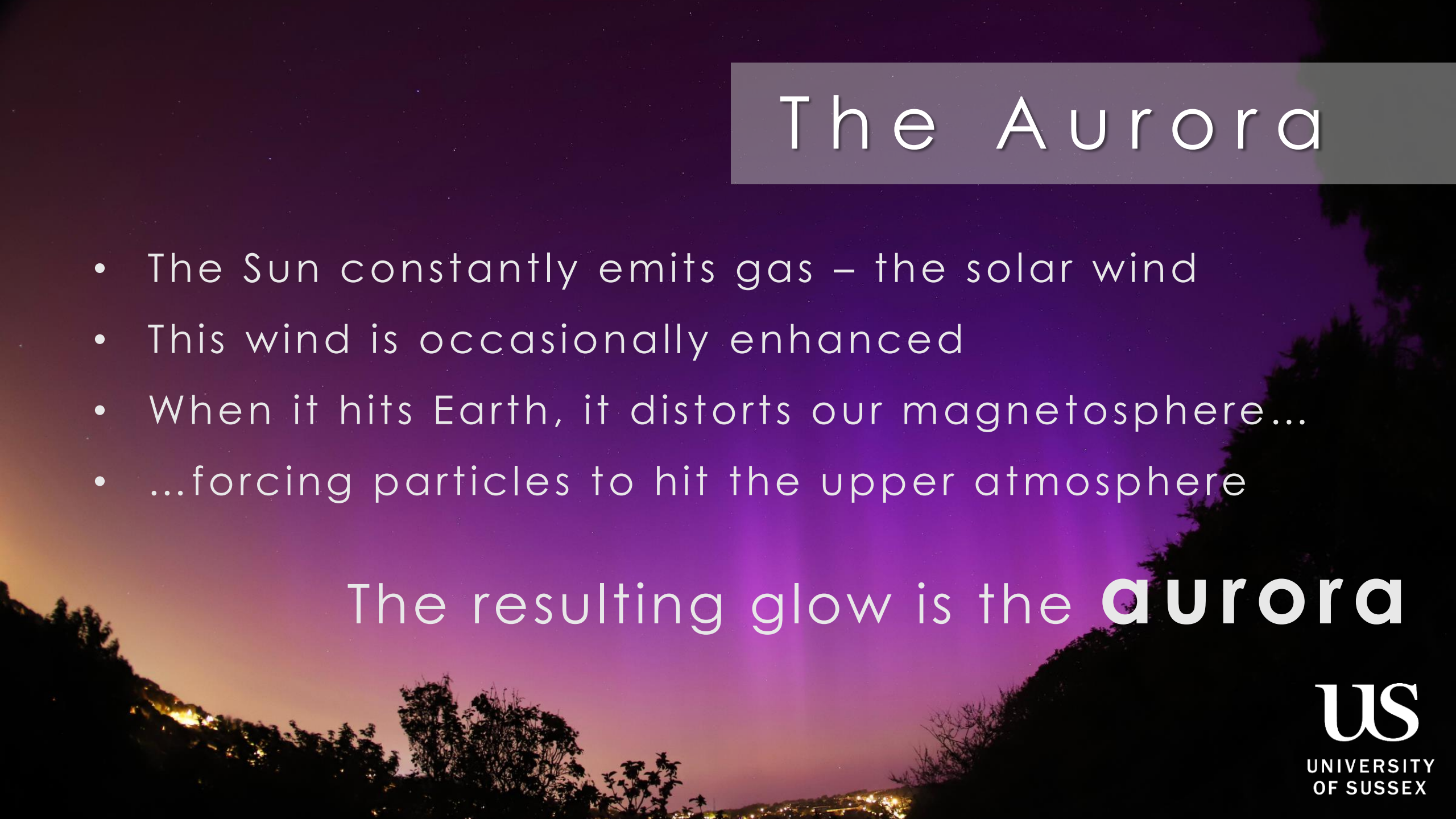
The Aurora

Dr Das Baskill
University of Sussex

The Aurora

- The Sun constantly emits gas – the solar wind
- This wind is occasionally enhanced
- When it hits Earth, it distorts our magnetosphere…
- …forcing particles to hit the upper atmosphere

The resulting glow is the **aurora**



UNIVERSITY OF SUSSEX





The Sun



White light photograph of the Sun, Saturday 17th October, 2026

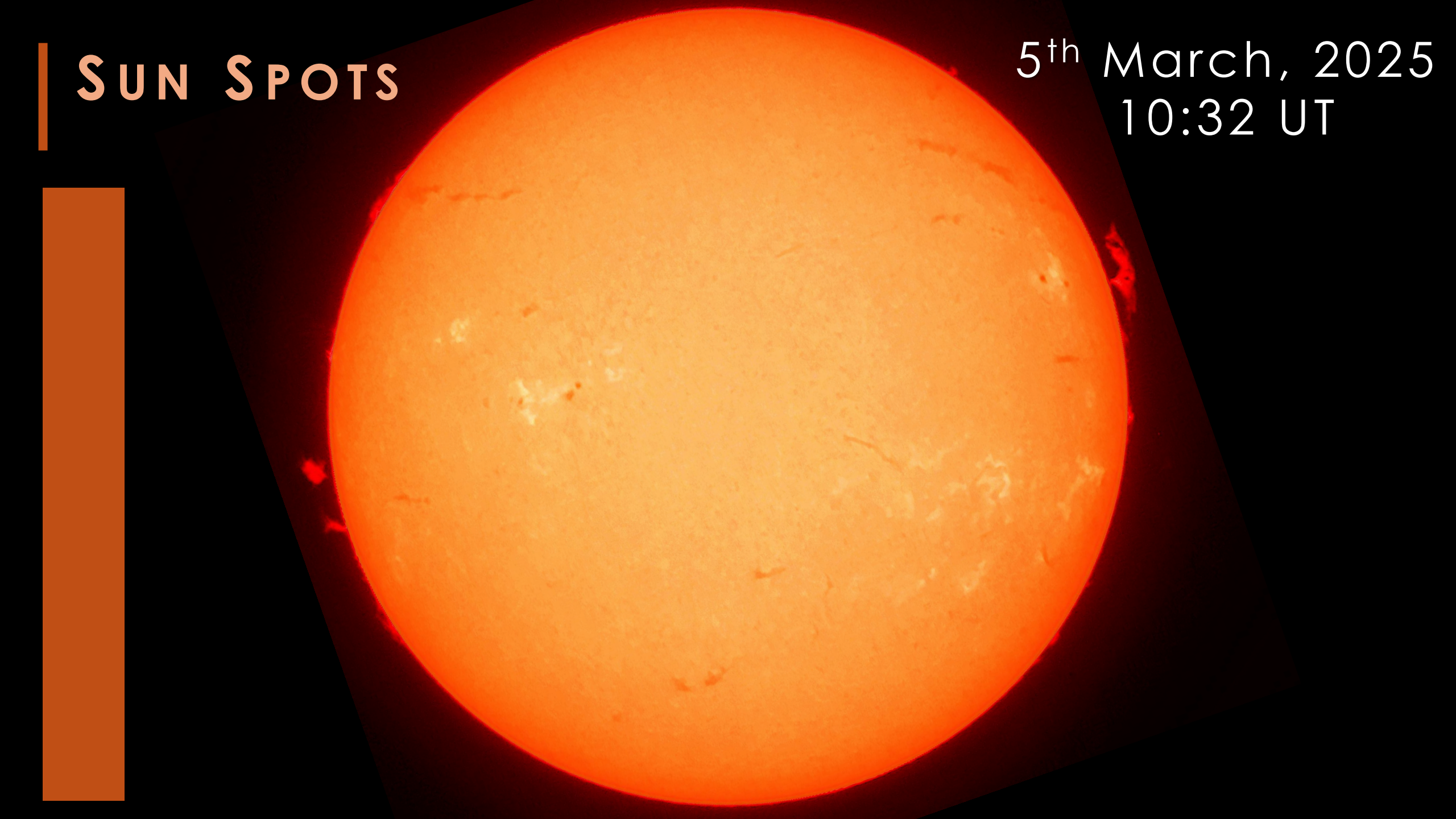
The Sun



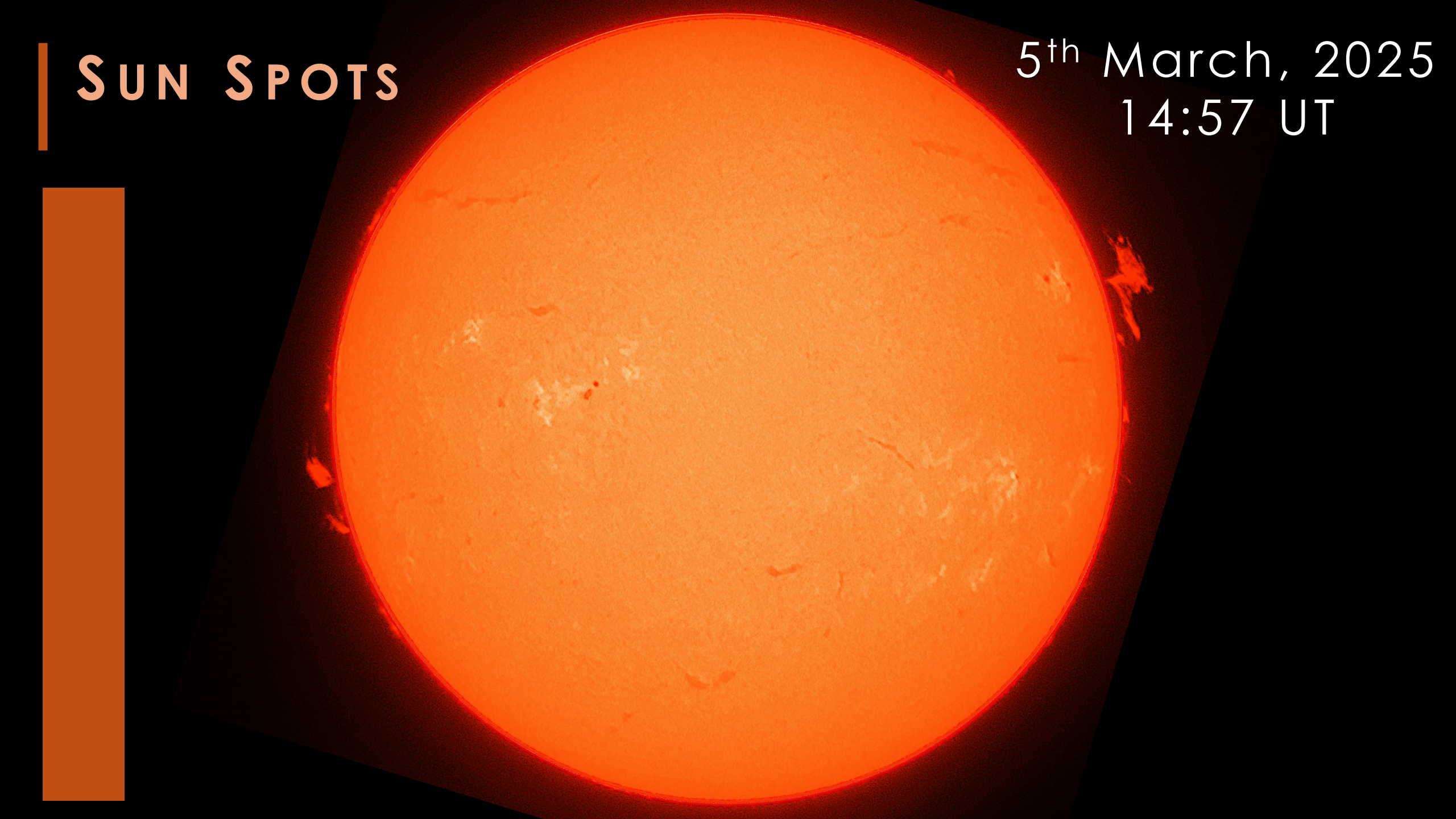
H-alpha view
of the Sun
4th August, 2024

Sun Spots

5th March, 2025
10:32 UT



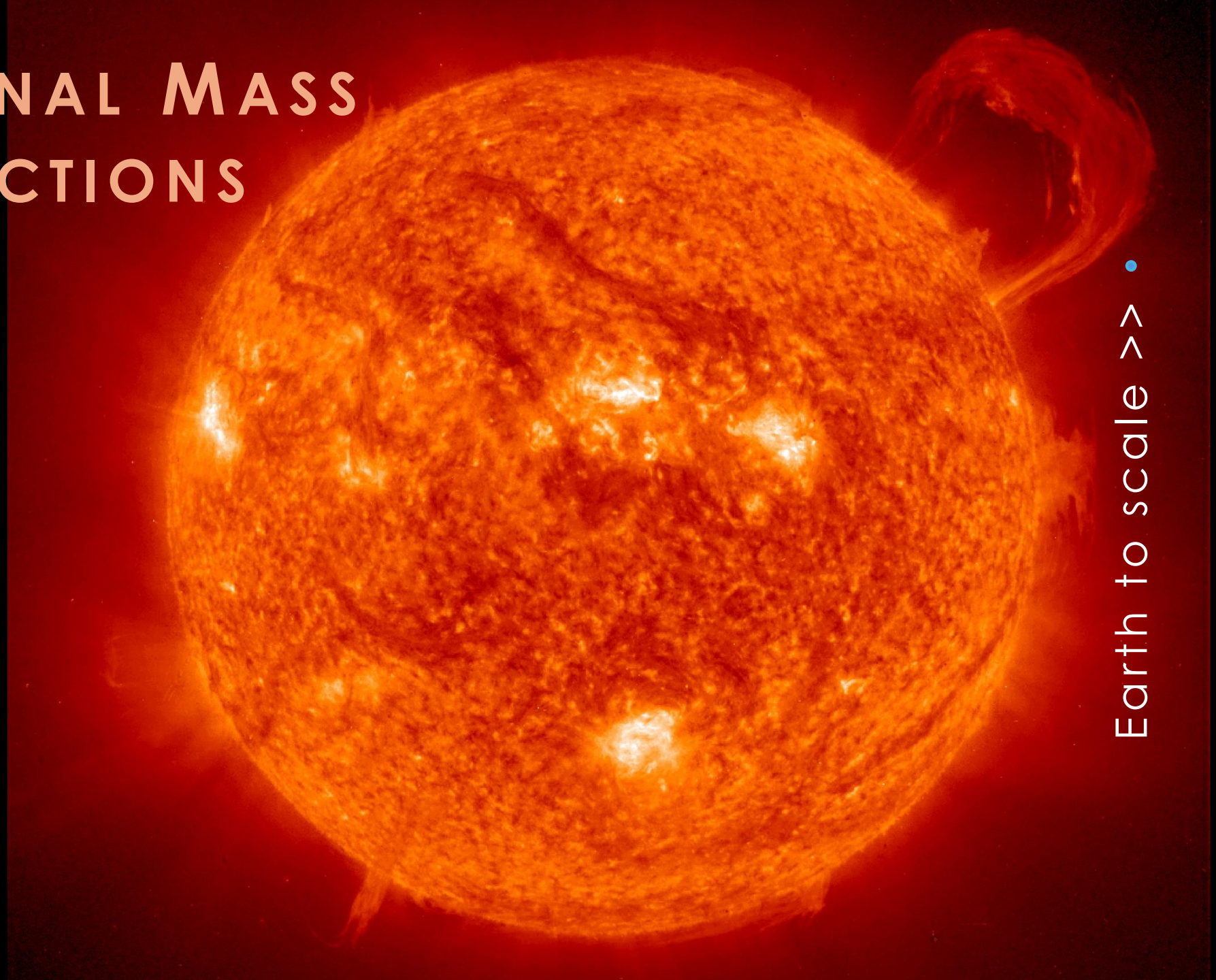
Sun Spots



5th March, 2025
14:57 UT



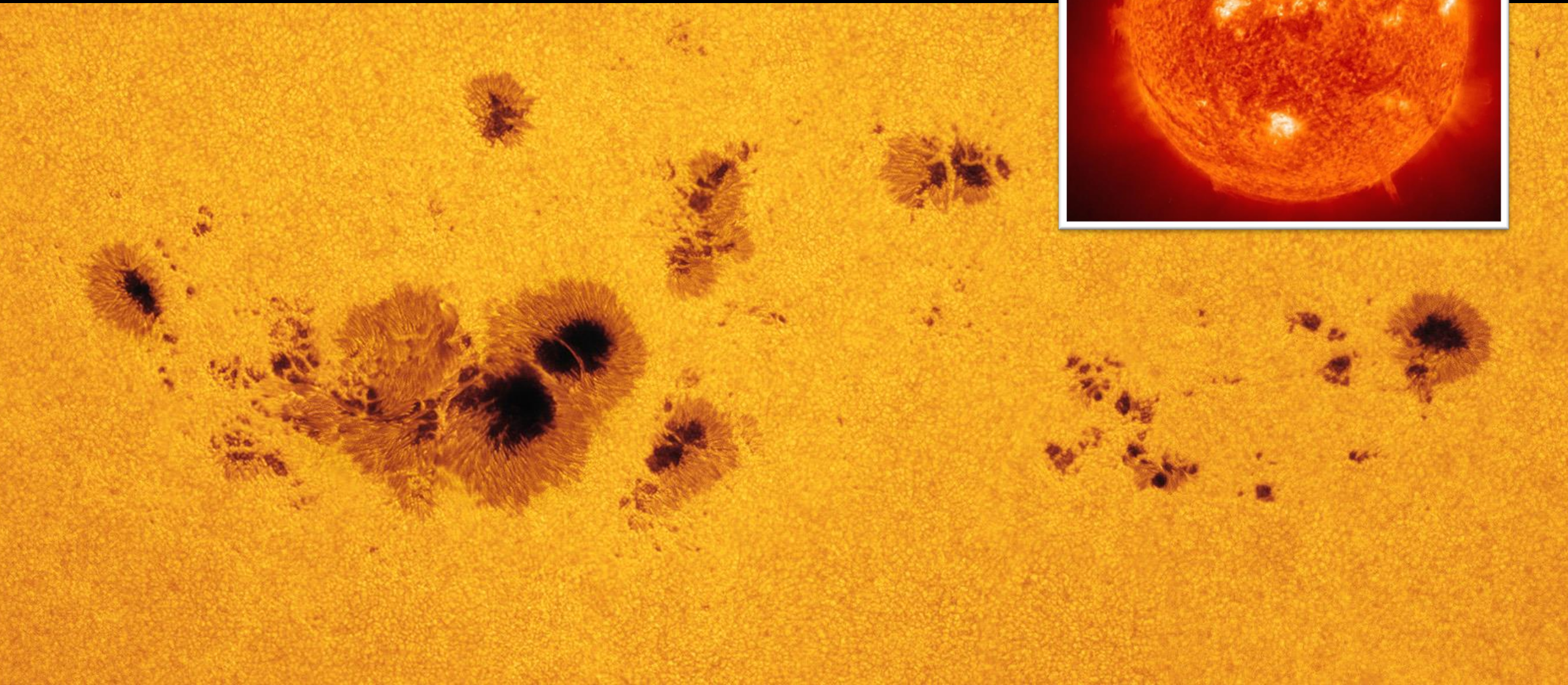
Coronal Mass Ejections

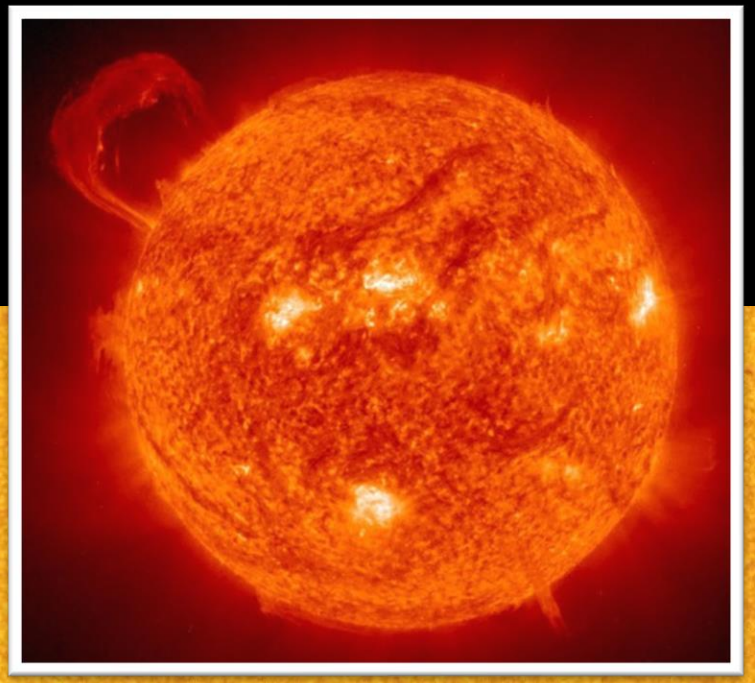


Earth to scale >>

SOHO Solar Observatory (Nasa/Esa)

Sun Spots

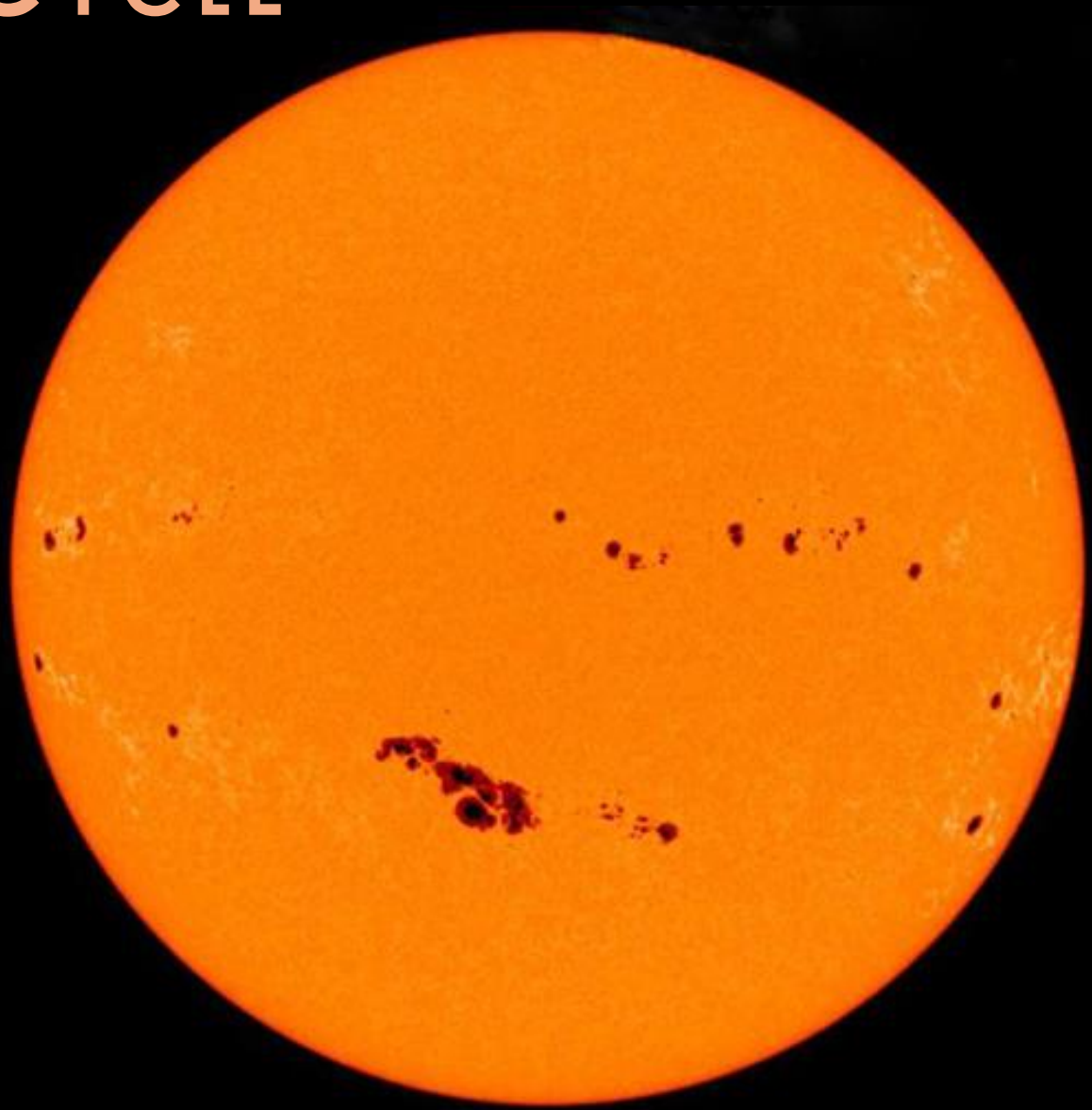




The 11-year Solar Cycle

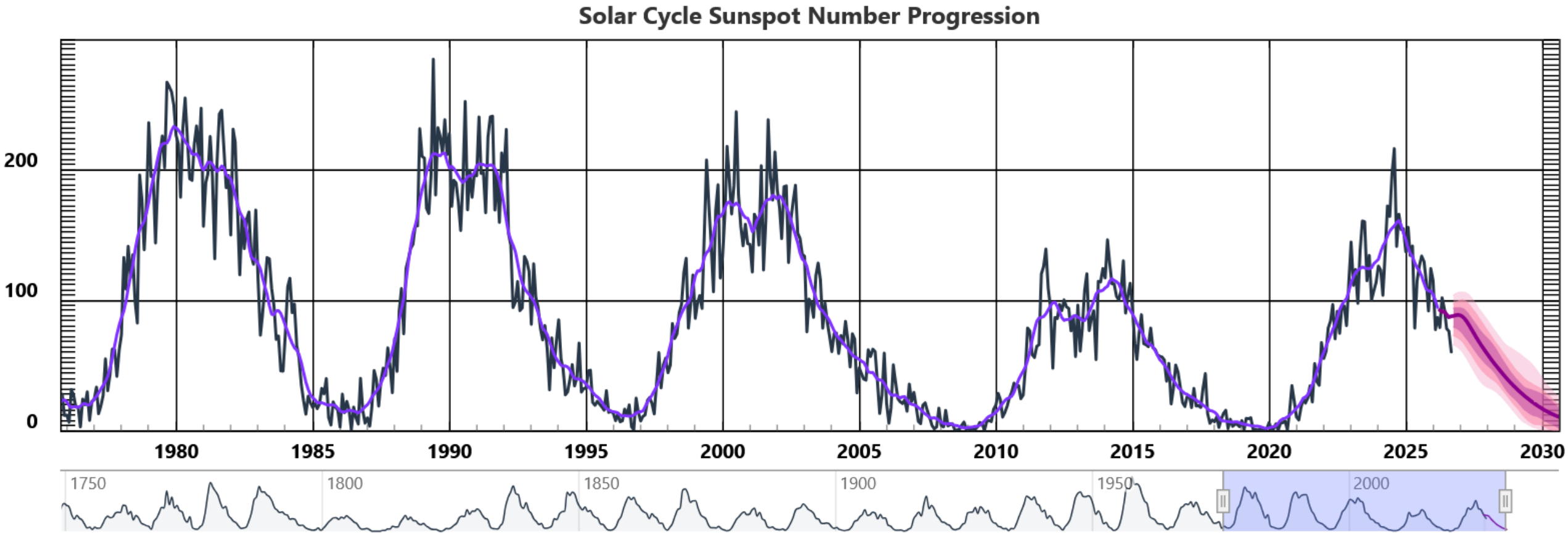


Solar Minimum



Solar Maximum

Solar Cycle


Solar Cycle Sunspot Number Progression
200
100
0
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
2030
1750
1800
1850
1900
1950
2000

Sun Spots

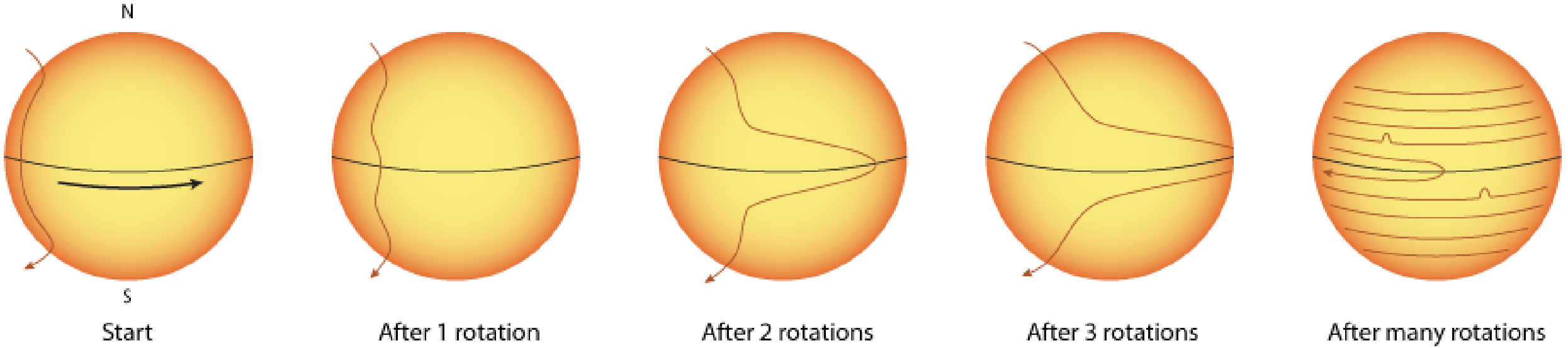
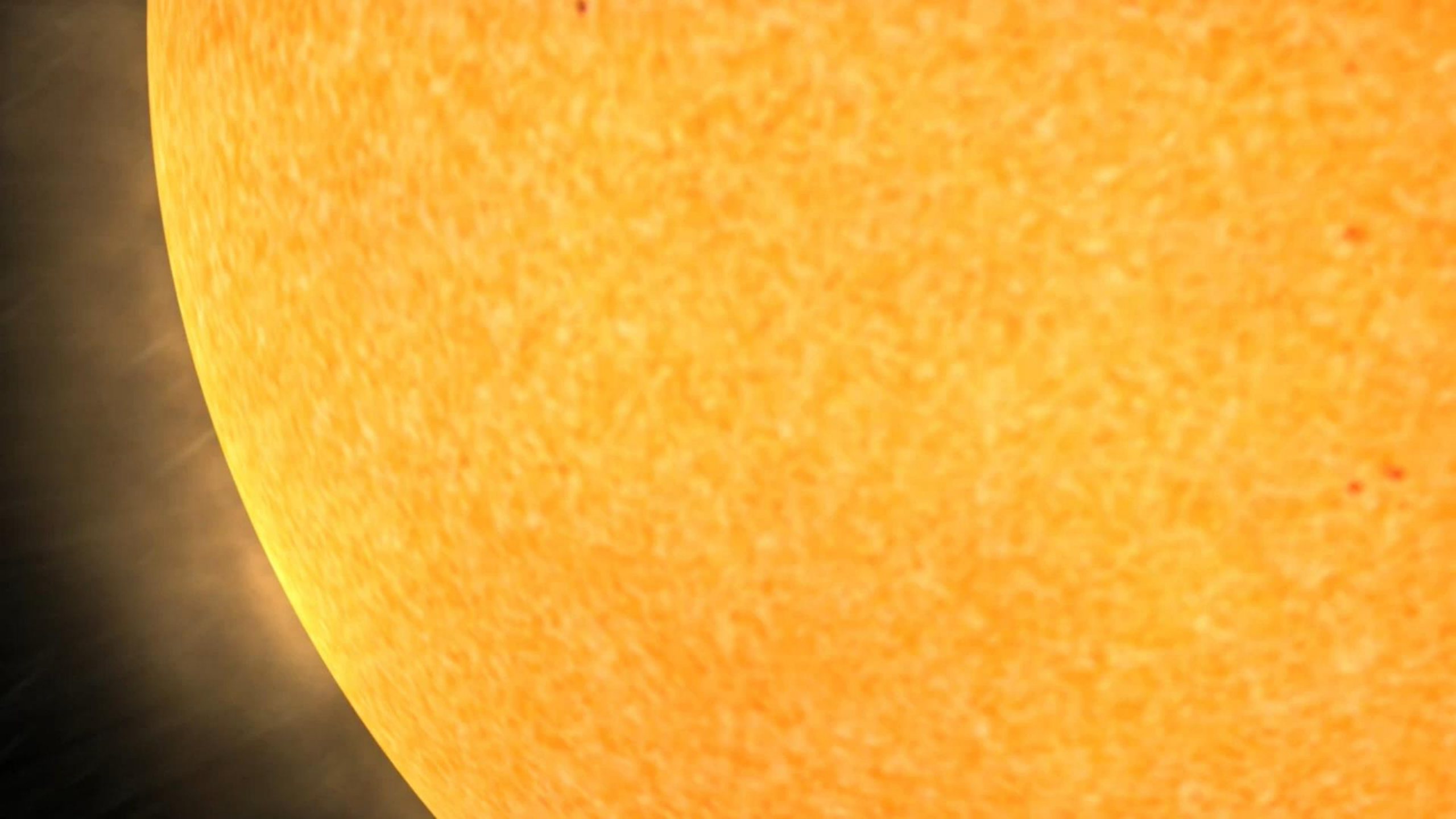
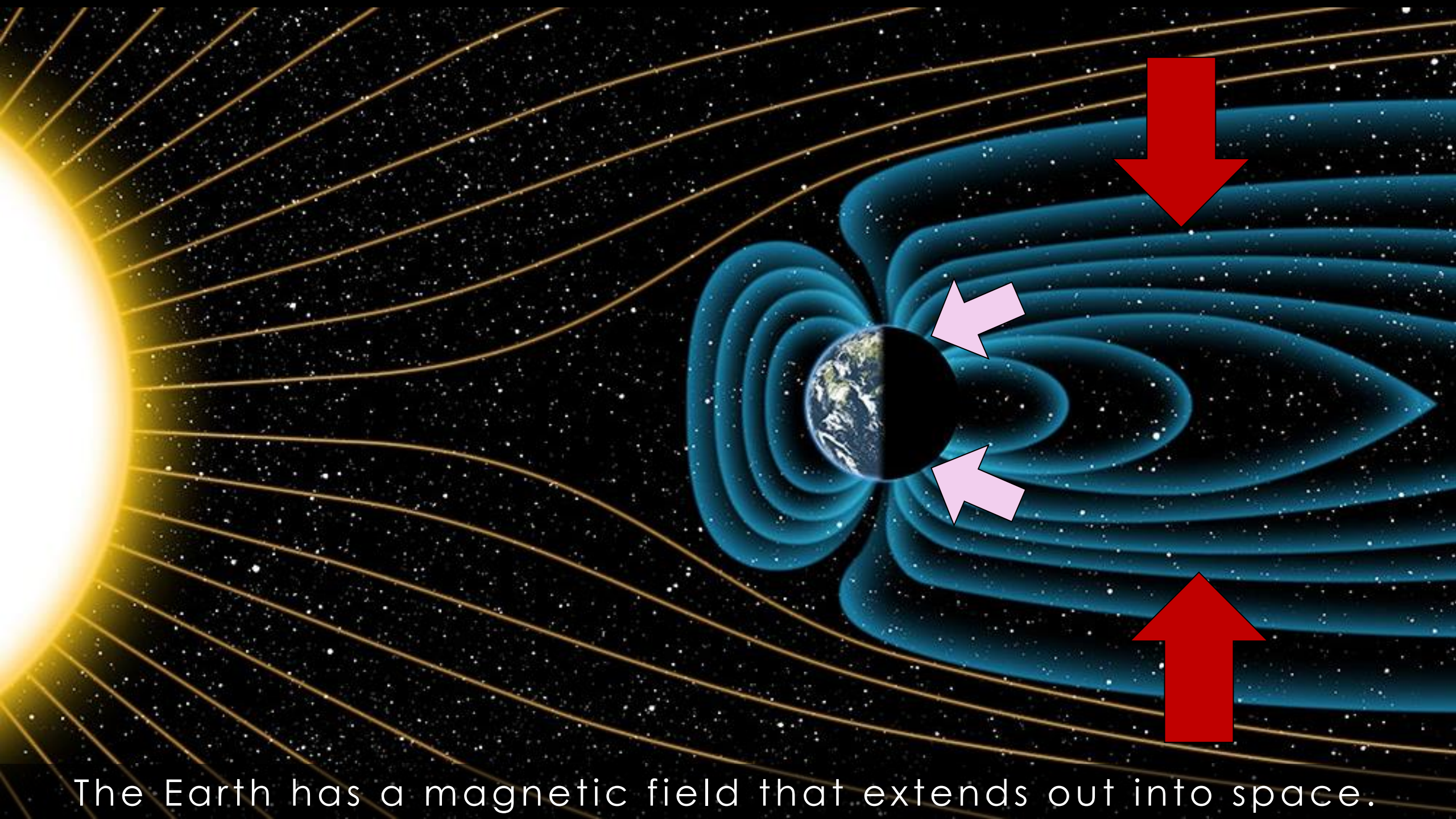
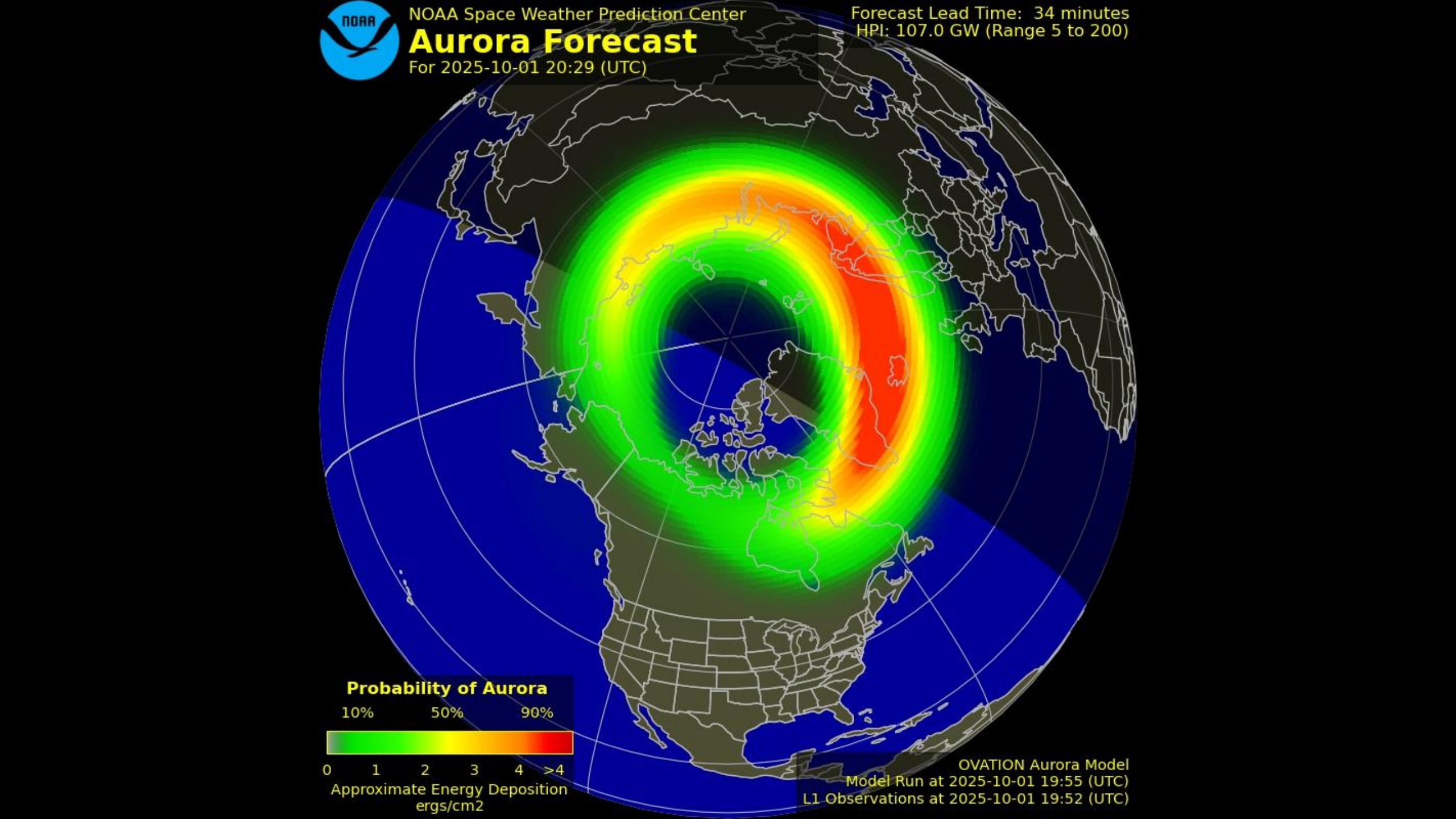


Illustration by José Francisco Salgado, PhD (Adler Planetarium)





The Earth has a magnetic field that extends out into space.


NOAA Space Weather Prediction Center
Aurora Forecast
For 2025-10-01 20:29 (UTC)
Forecast Lead Time: 34 minutes
HPI: 107.0 GW (Range 5 to 200)
Probability of Aurora
10%
50%
90%
0
1
2
3
4
>4
Approximate Energy Deposition
ergs/cm2
OVATION Aurora Model
Model Run at 2025-10-01 19:55 (UTC)
L1 Observations at 2025-10-01 19:52 (UTC)


August 2014 over Greenland

The Colours of the aurora



Red

High-altitude oxygen >200km, visible during strong solar activity.

Green

Common due to the abundance of Oxygen between 100-200 km.

Pink, purples & blues

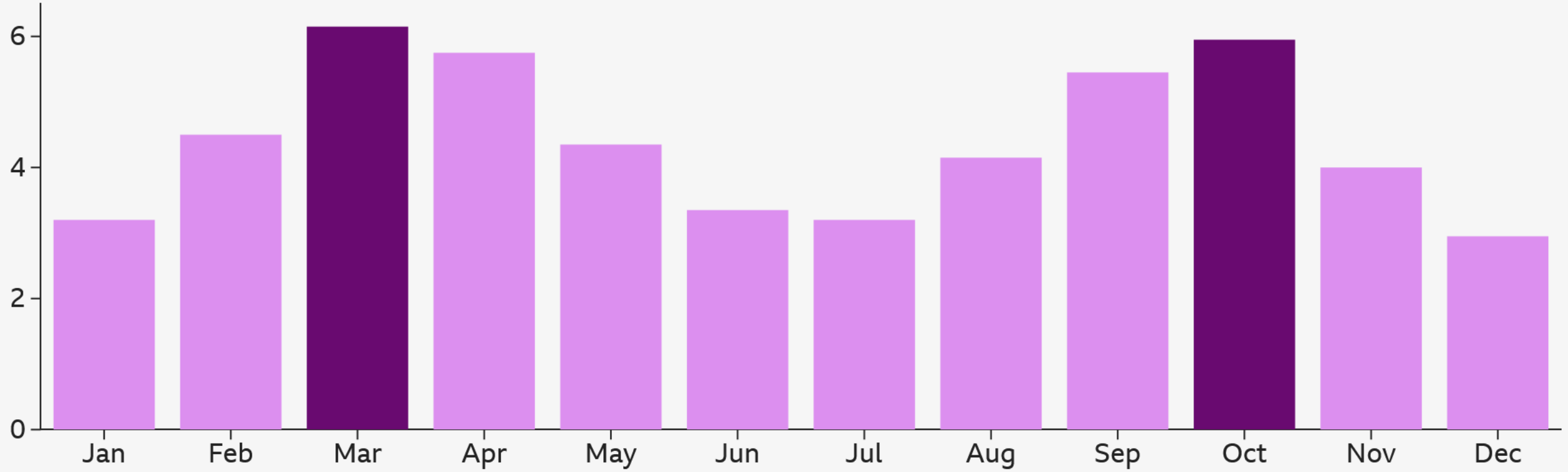
Nitrogen molecules emit red and blue colours when energetic particles reach lower altitude.



Stronger aurora more likely in spring and autumn

Effects are caused by 'geometric disturbances' - which increase around **equinox months**

Average number of geometrically disturbed days each month 1932-2007



Source: Nasa physicist David Hathaway

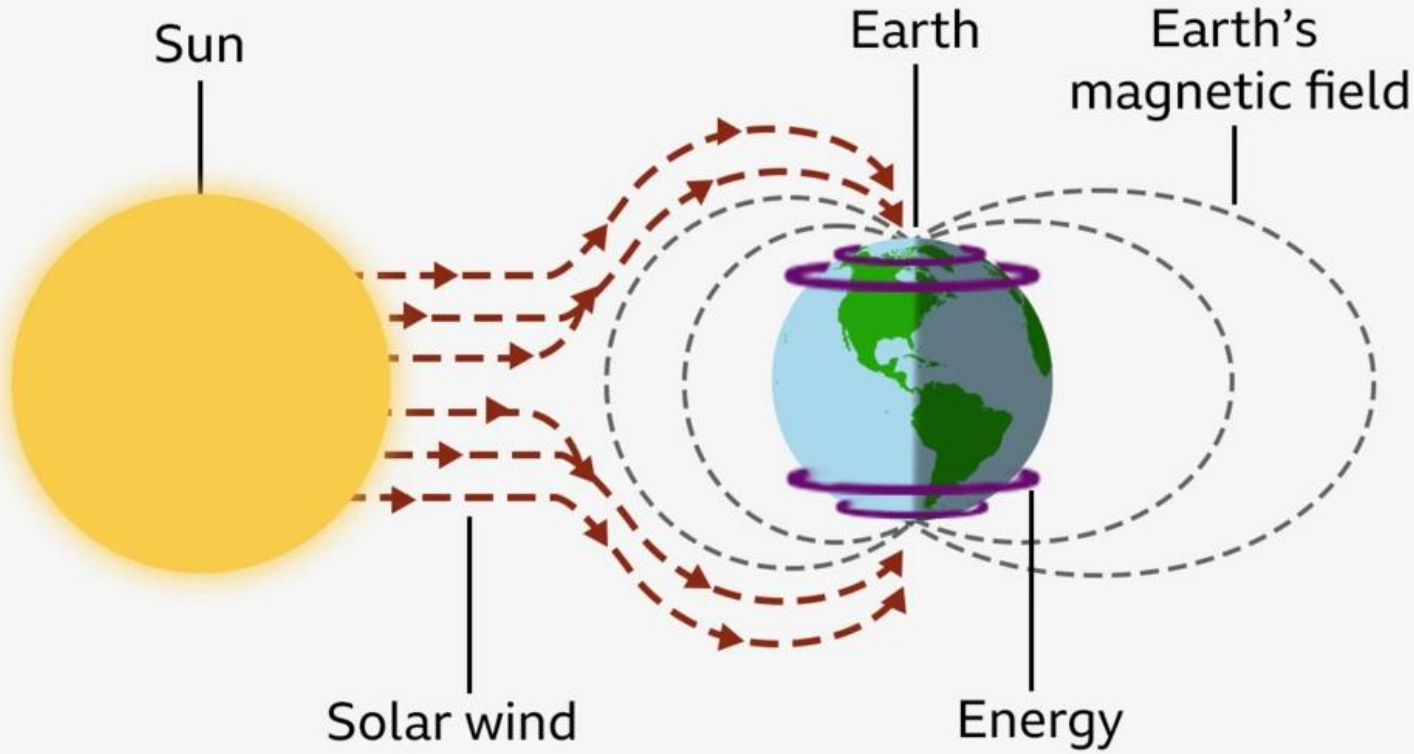
BBC

Russell-McPherron effect, due to the Earth's tilt

Equinoxes

How the Russell-McPherron effect can make the aurora stronger

When there is no tilt - at **equinox -** the magnetic field of the solar wind is more attracted to the Earth's magnetic field - resulting in more energy entering the atmosphere and **stronger aurora**



Source: BBC research/Noaa

BBC

WEBSITES

https://www.gi.alaska.edu/monitors/aurora-forecast

AURORA FORECAST

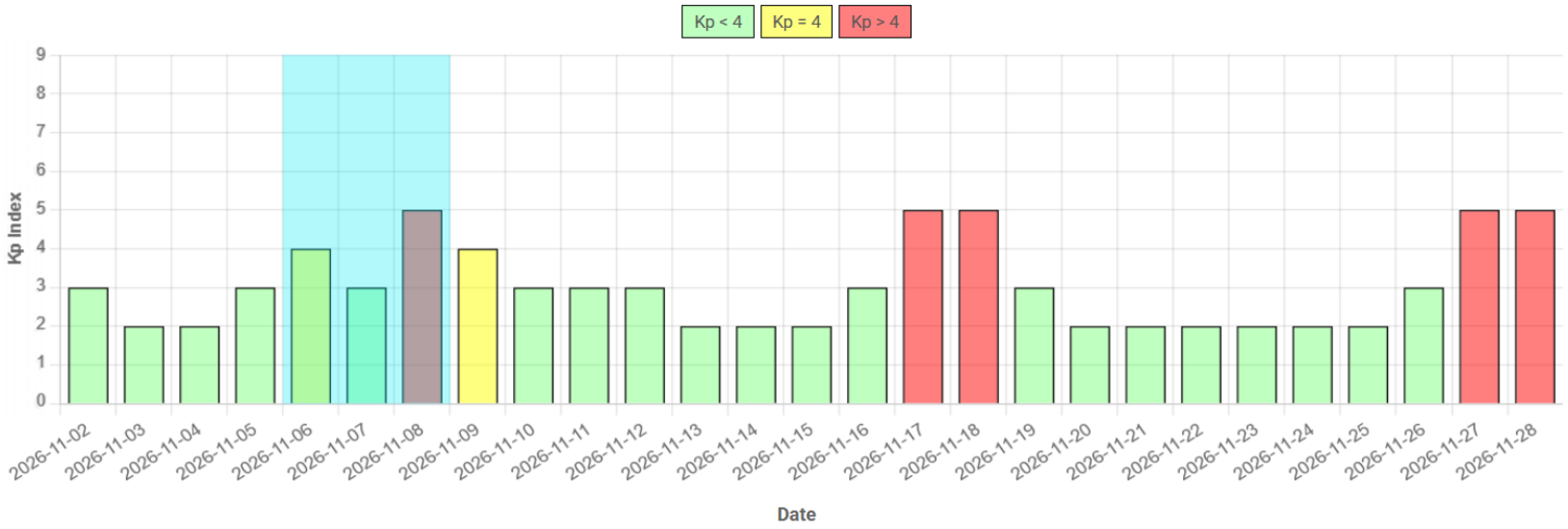
< Previous Day | Current | Next Day >

3-Day Forecast — More Reliable

27-Day Forecast — Less Reliable

Historical Data — Actual Activity

Time Zone: Local (Greenwich)



Data source: We pull data from the NOAA 27-Day Space Weather Outlook every Monday at 16:00 PM (Greenwich). The Space Weather Outlook is the forecast of geomagnetic activity in 27 days. NOAA updates the 27-Day Space Weather Outlook every Monday at 15:00 UTC.

The data: Highlighted in blue are 3-Day Forecast data overlaying the 27-Day Forecast. 3-Day Forecast data are updated daily and provides a more accurate forecast.

Last updated: 2026-11-02 16:00 PM (Greenwich).

FAQ

Read our FAQs to learn about these common questions and more:

Viewing the Aurora

- What time of year is best for seeing aurora?
- Can I see aurora during the summer months?
- Where is the best place to see aurora?
- Where should I view the aurora when I'm in Fairbanks, AK?

Science of the Aurora

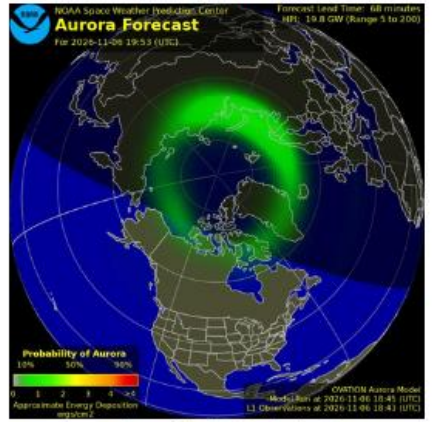
- What are Kp numbers?
- What does the 2024-25 solar maximum mean for aurora?
- Can you predict where there will be aurora?

Forecasting the Aurora

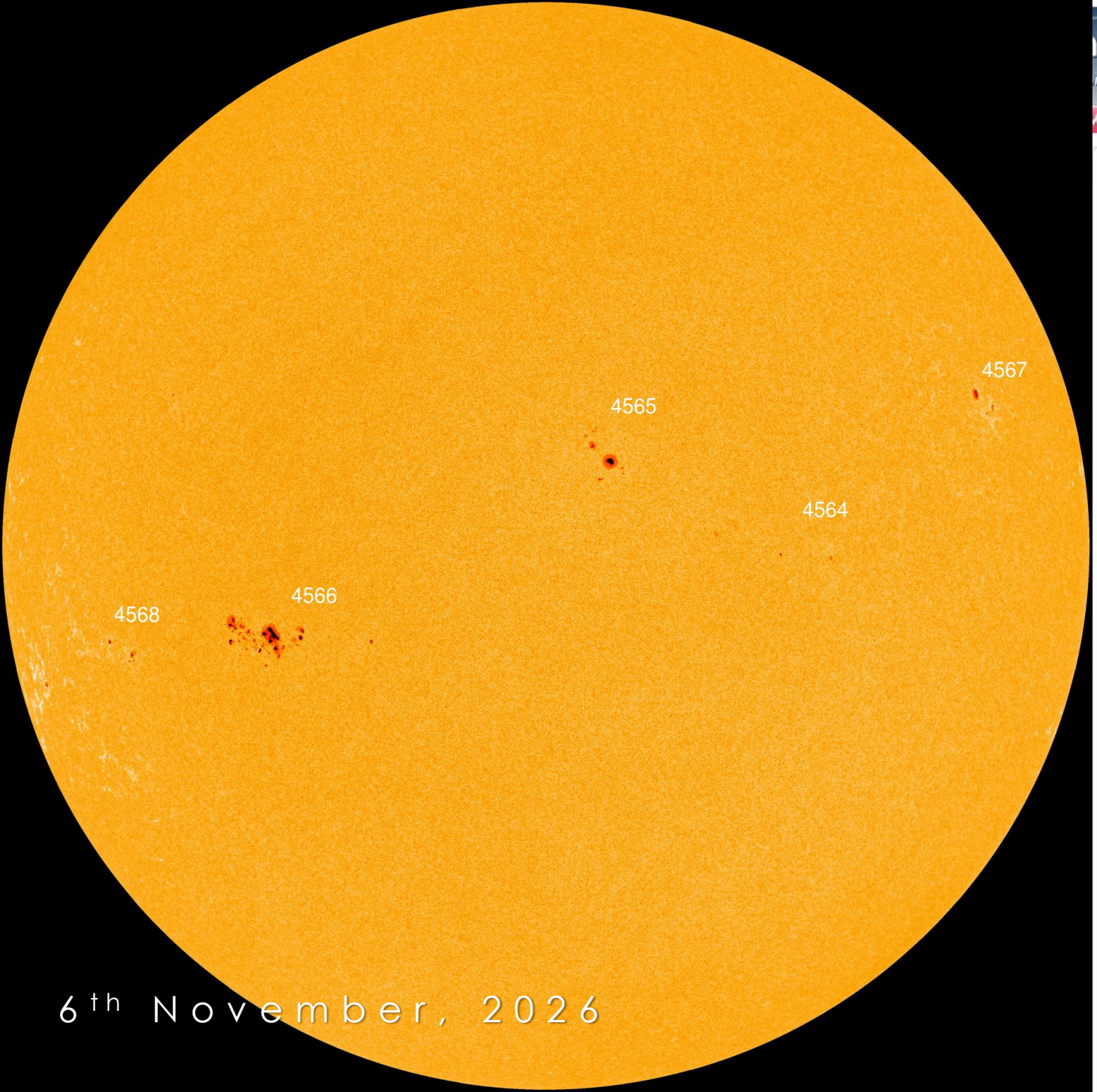
- How accurate is the 3-day forecast?

Current Activity

Watch the last few seconds of the animation to see what is predicted over your location in the next 30 minutes.



Source: Space Weather Prediction Center



er.com
n about the Sun Earth environment
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IT YOUR PHOTOS | CONTACT US | SUBSCRIBE | FLYBYS | EARTH TO SKY

What's up in space

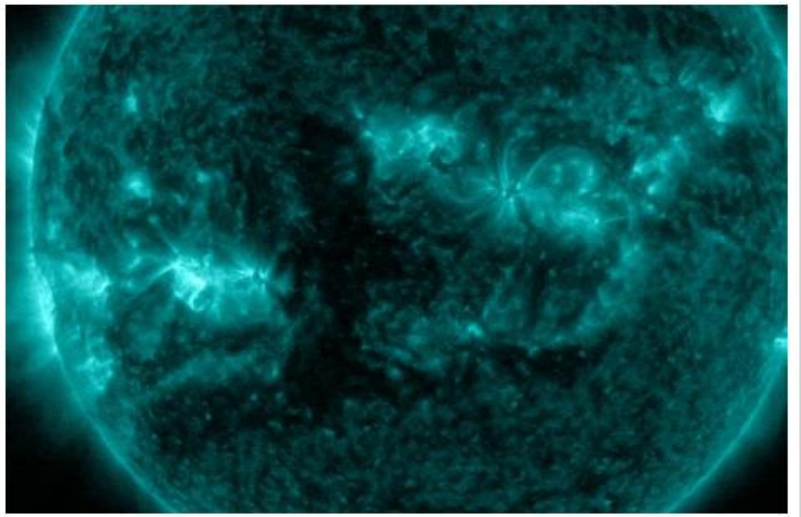
Friday, Nov. 6, 2026

AI writing is everywhere -- except here. This website is written by Dr. Tony Phillips, a carbon-based lifeform with 30 yrs of forecasting experience. Our guarantee: If you find a mistake, it was made by a real human being.



A HOLE IN THE SUN'S ATMOSPHERE: Look at this. Do you see a witch flying on a broom? Actually, it is a hole in the sun's stmosphere sending a stream of solar wind toward Earth. First contact on **Nov. 7th** could spark G1/G2-class geomagnetic storms and high-latitude auroras. **Aurora alerts:** SMS Text

POTENTIALLY DANGEROUS SUNSPOT: An active sunspot (AR4566) has emerged in the sun's southern hemisphere. Yesterday, it produced its first significant solar flare--an M3.5-class eruption recorded by NASA's Solar Dynamics Observatory:



The explosion launched a faint CME into space, just outside the strike zone for Earth. It will miss our planet.

The next explosion might be stronger and better aimed. AR4566 has a 'beta-gamma-delta' magnetic field that harbors energy for X-class solar flares, and it is turning toward Earth. Geoeffective explosions are possible this week. Solar

archives

November
6
2026
view

SATELLITE DRAG

today: **54.9** m/day ↑



2026 max: **127.6** (Jan 20)
2026 min: **19.7** (Aug 10)

Explanation: This is how fast a typical Starlink at 480 km is sinking due to aerodynamic drag: more information.
Updated: 6 Nov 2026

Starlink Statistics
Total Launched: 13,069
In orbit: 11,214
Updated: 05 Nov 2026

Starlink Re-entries (2026)
January: 24
February: 54
March: 48
April: 51
May: 60
June: 35
July: 52
August: 32
September: 47
October: 46
November: 7
Updated: 05 Nov 2026

Starlink Re-entries (yearly)
2020 total: 46
2021 total: 78
2022 total: 100
2023 total: 88
2024 total: 308
2025 total: 657
2026 total: 445 so far

See the aurora borealis from the UK

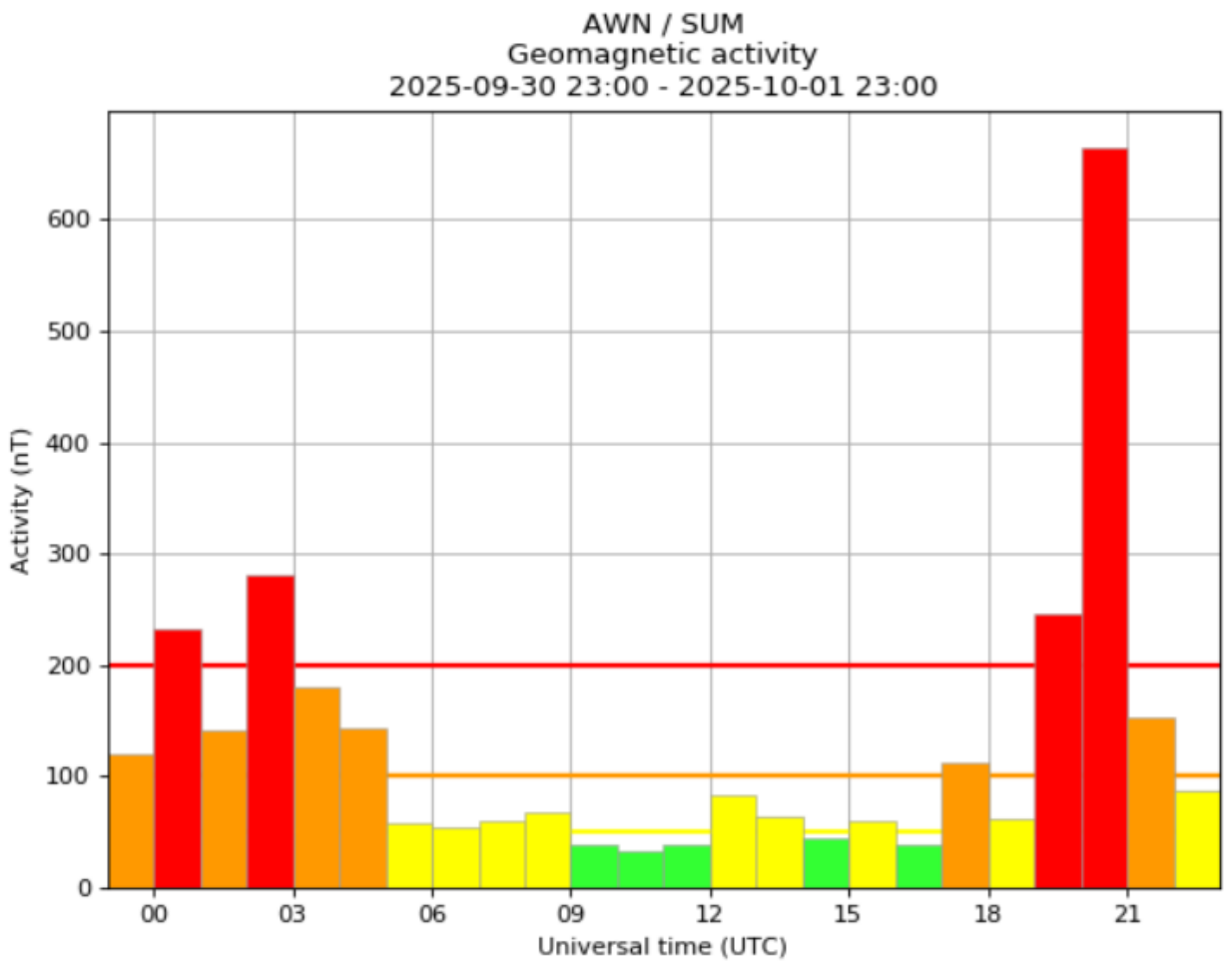
AuroraWatch UK

- Home
- Alerts
- Shetland alerts
- Plots
- Map
- Build
- FAQs
- Contact Us
- Blog
- Twitter
- Telegram
- Facebook
- Flickr
- Instagram
- YouTube

Lancaster University

AuroraWatch UK status:

Minor geomagnetic activity



Activity plot | Magnetogram | Stackplot

For Sussex...

500nT
– visible to cameras

750nT
– potentially visible by eye

1000nT
– easy to see by eye

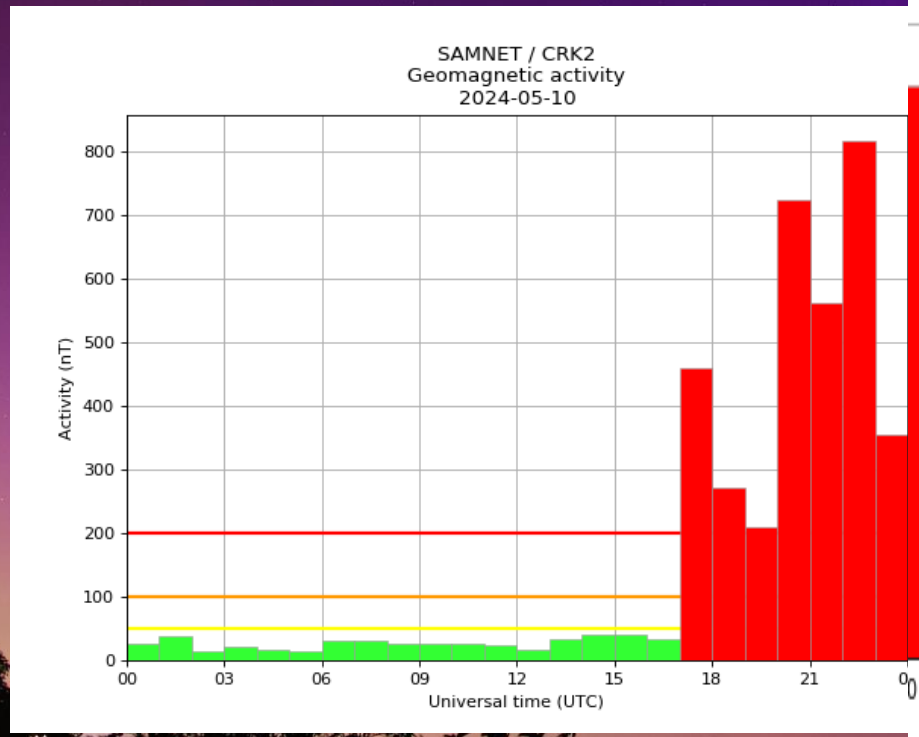
https://aurorawatch.lancs.ac.uk/

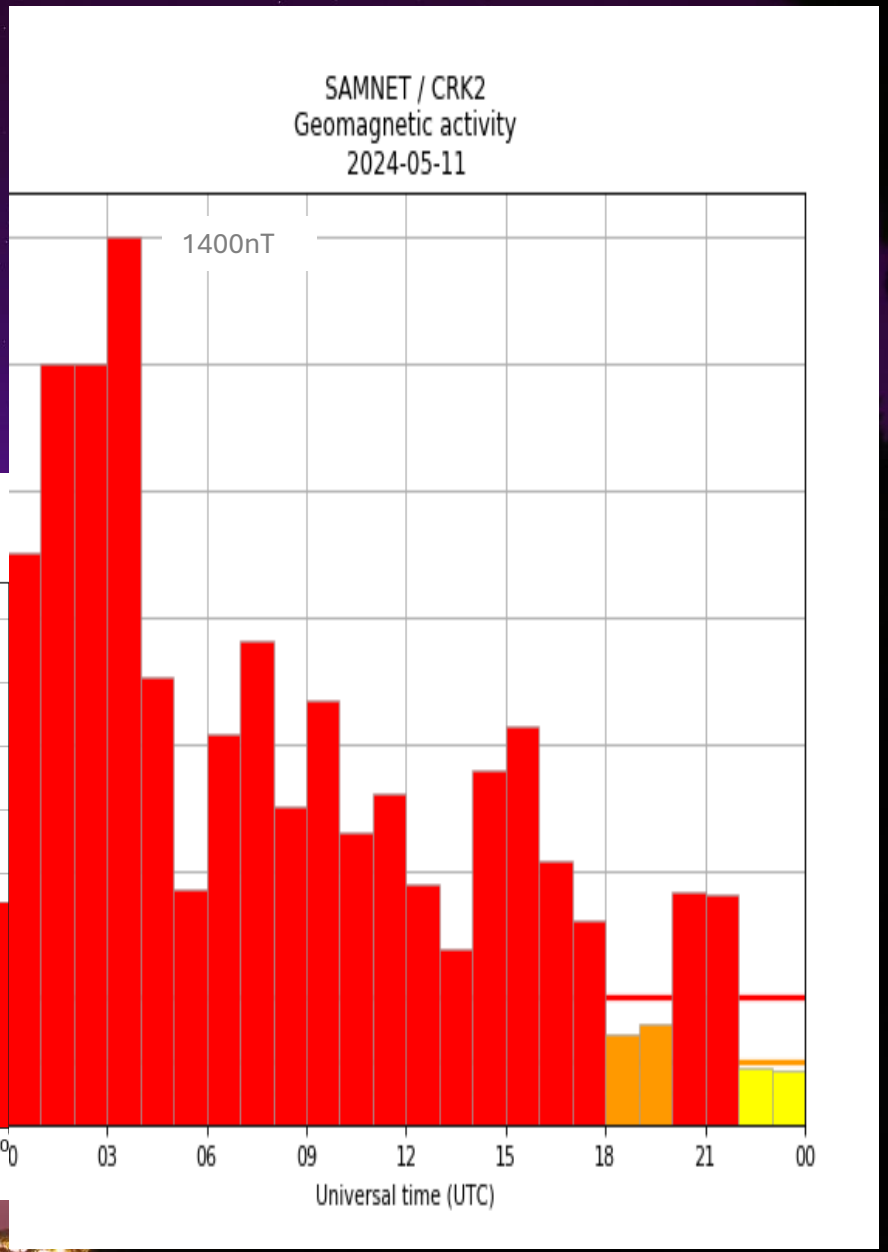
© AuroraWatch UK 2015.

Recent high aurora activity



Brighton – May 2024





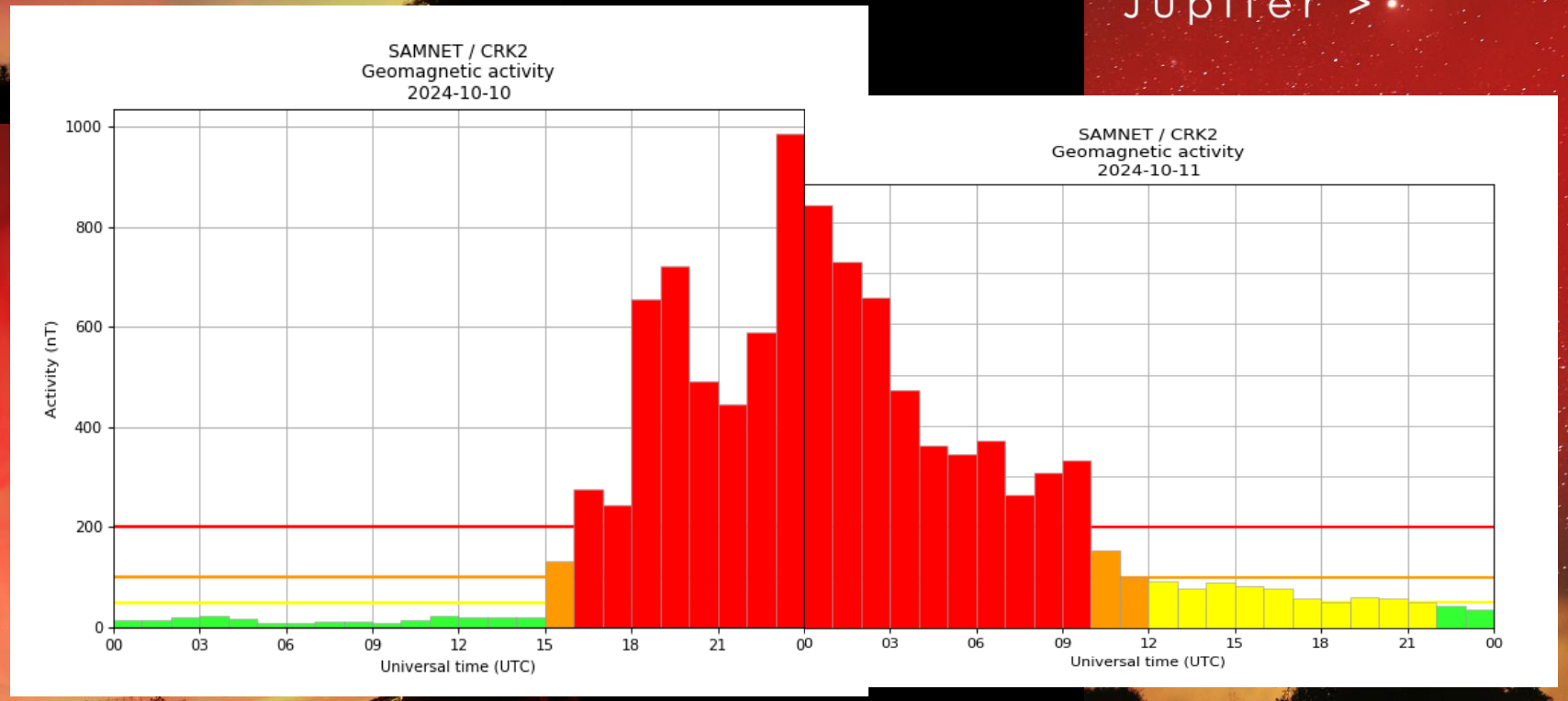
Brighton – October 2024

Pleiades >

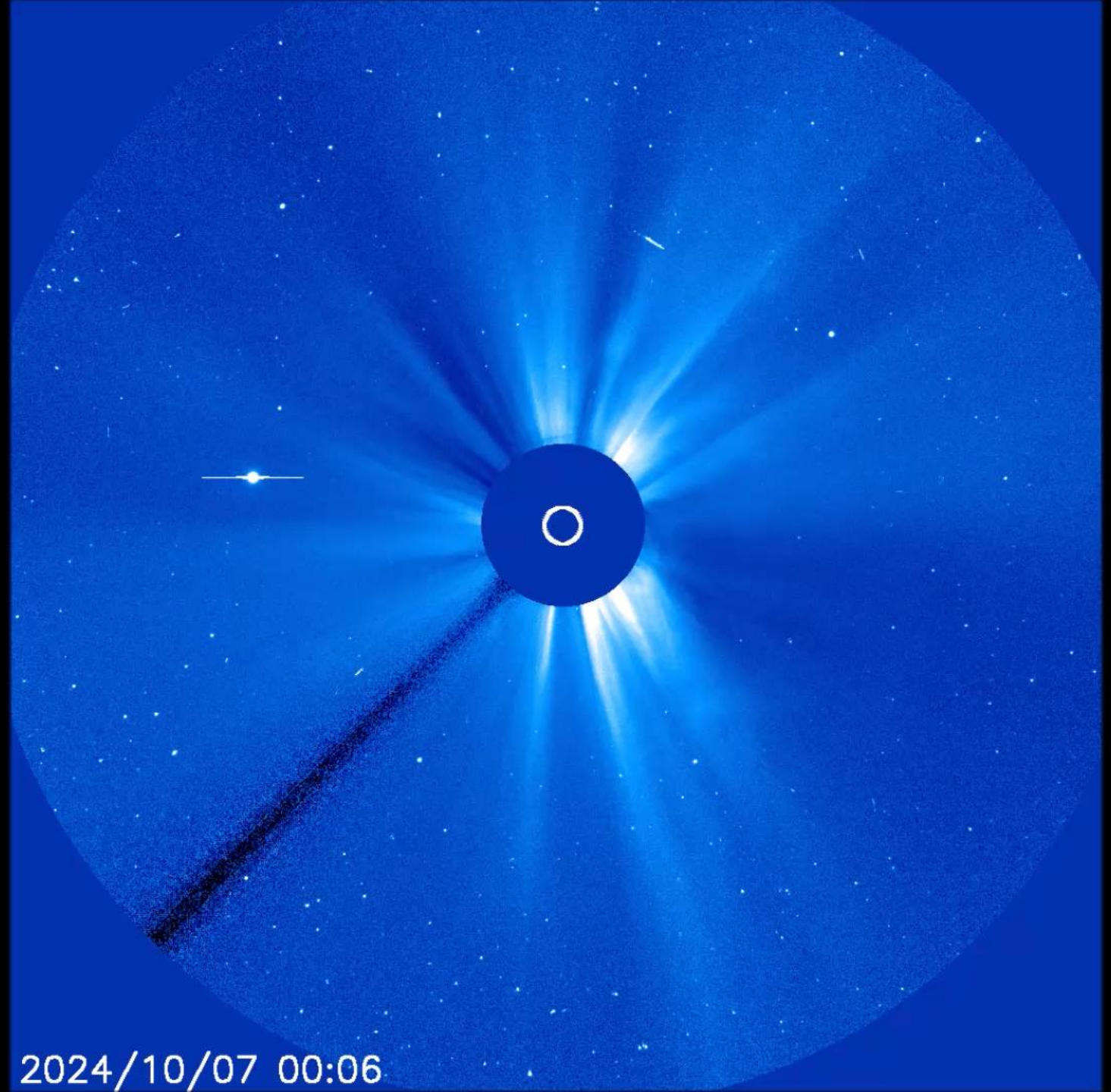
Hyades

Jupiter >

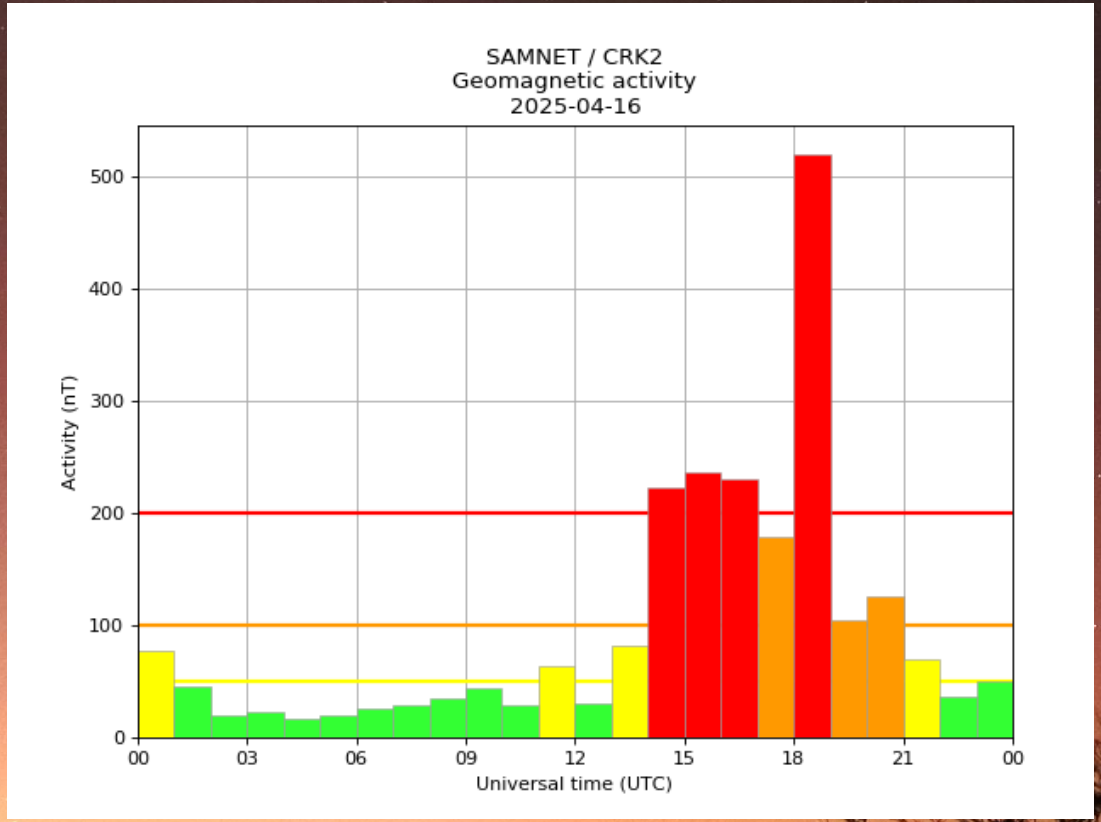
Orion



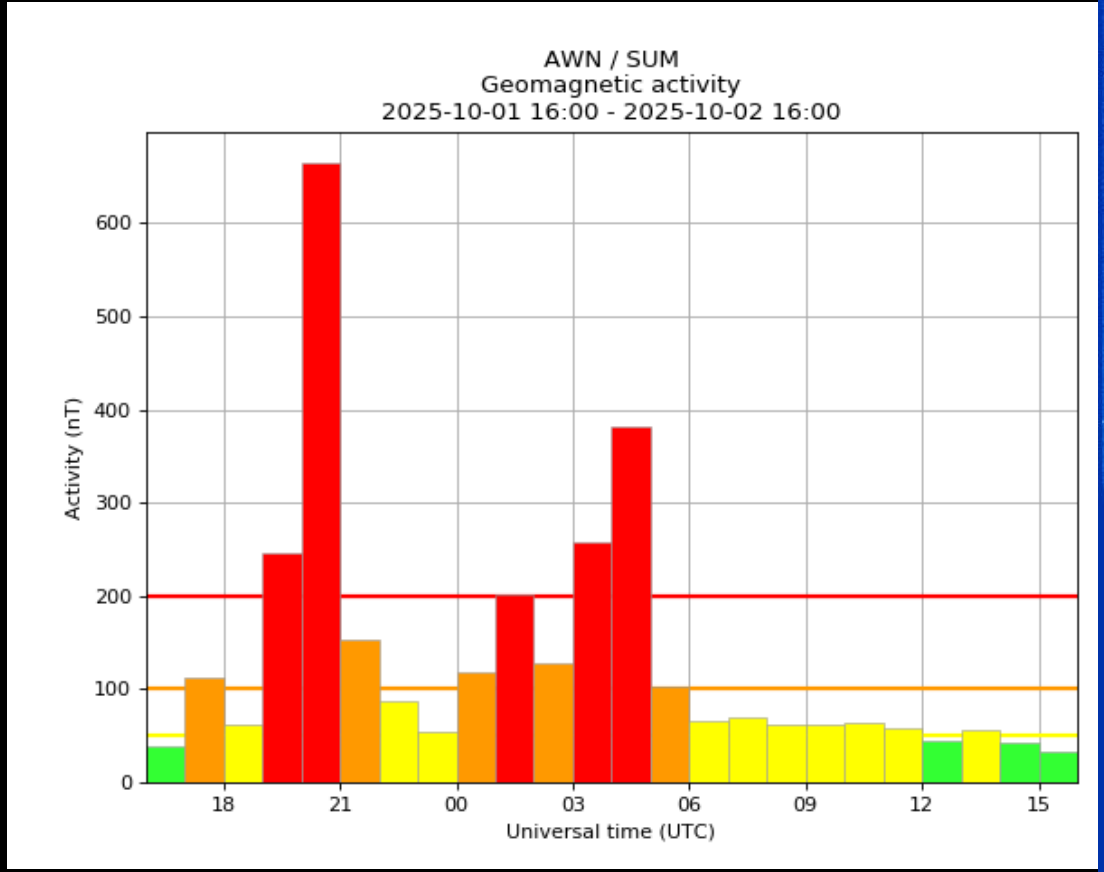
The Solar activity that sparked aurora over southern England in October 2024

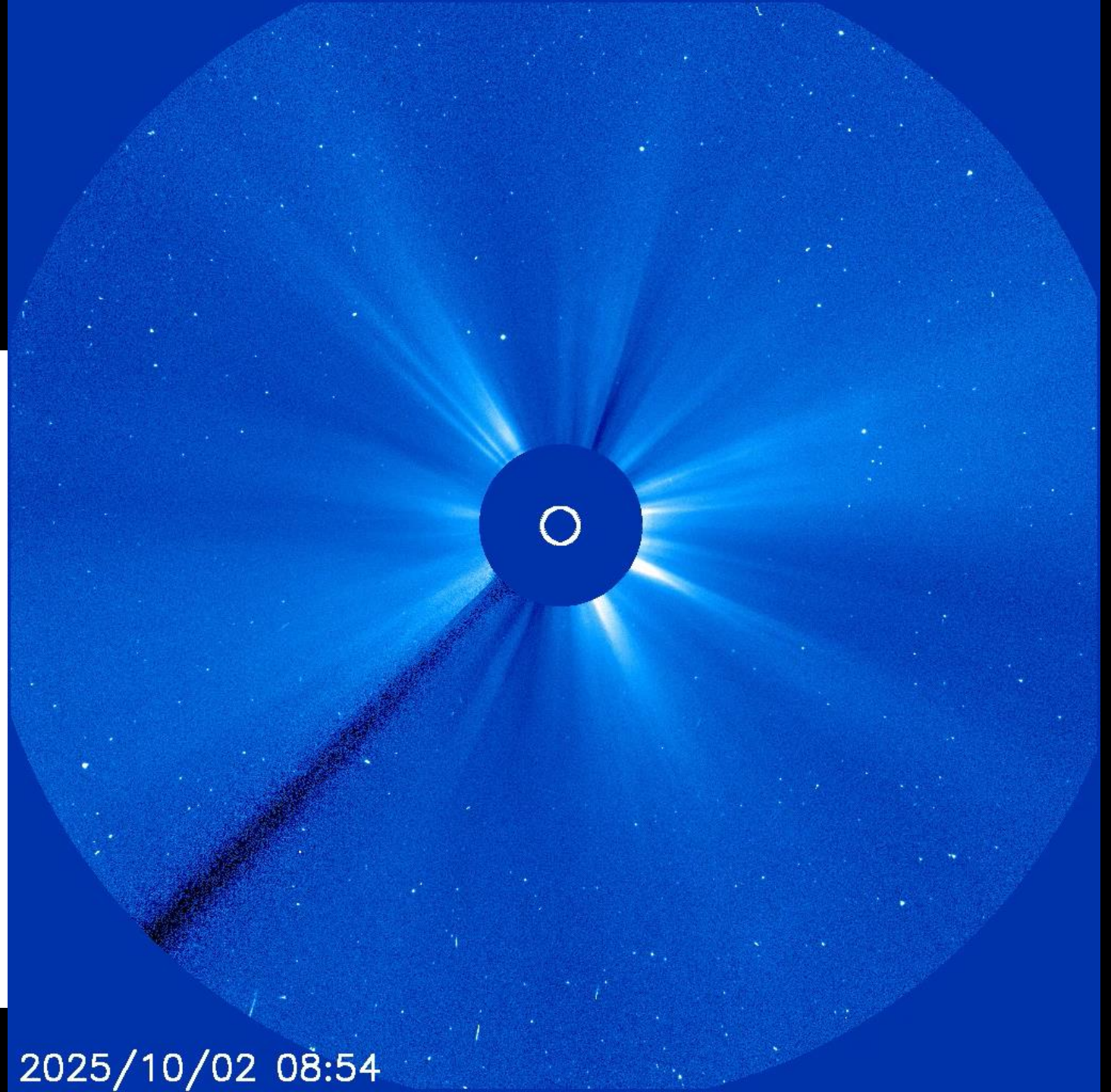


Brighton – April 2025

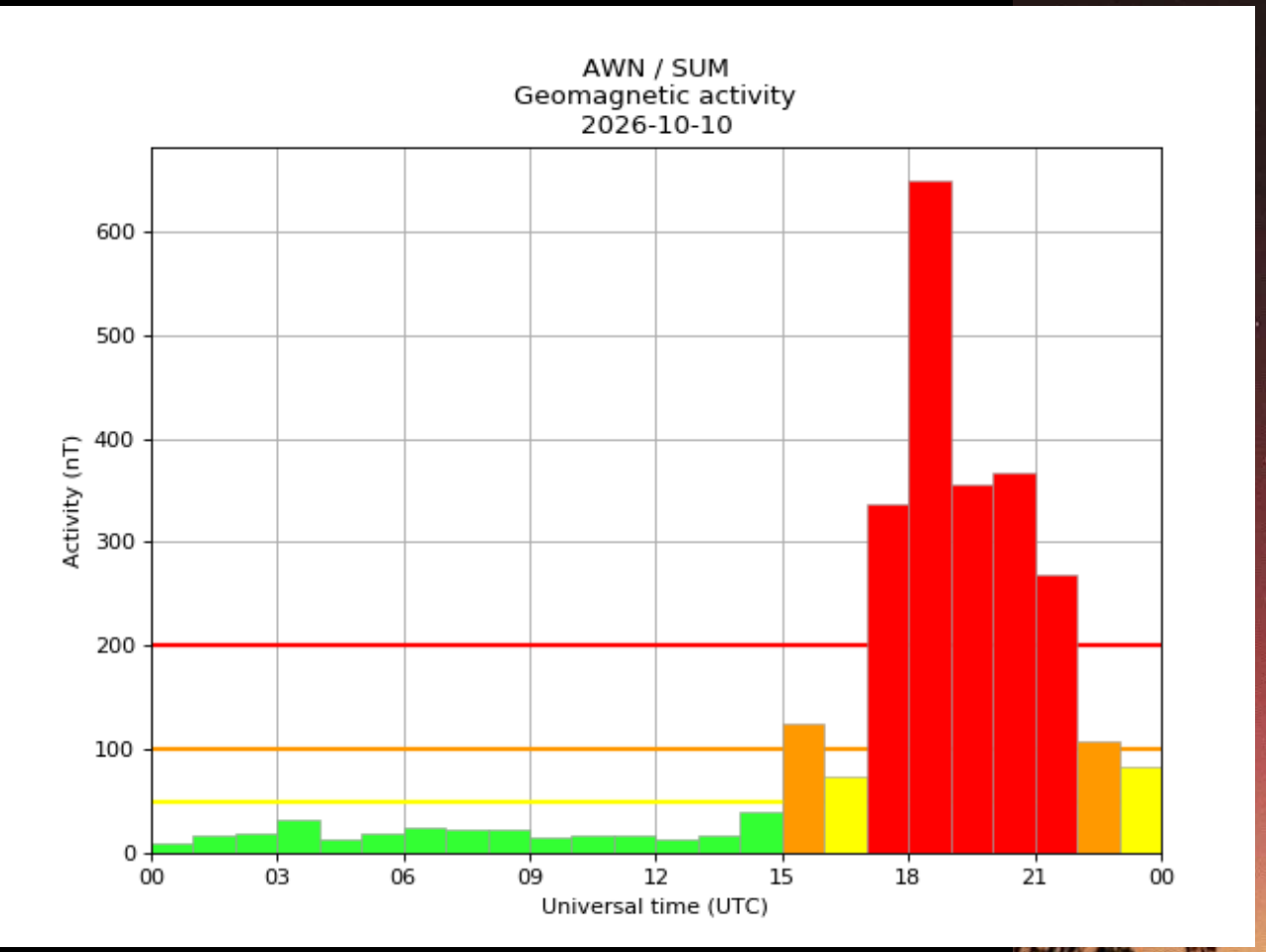


Brighton – October 2025

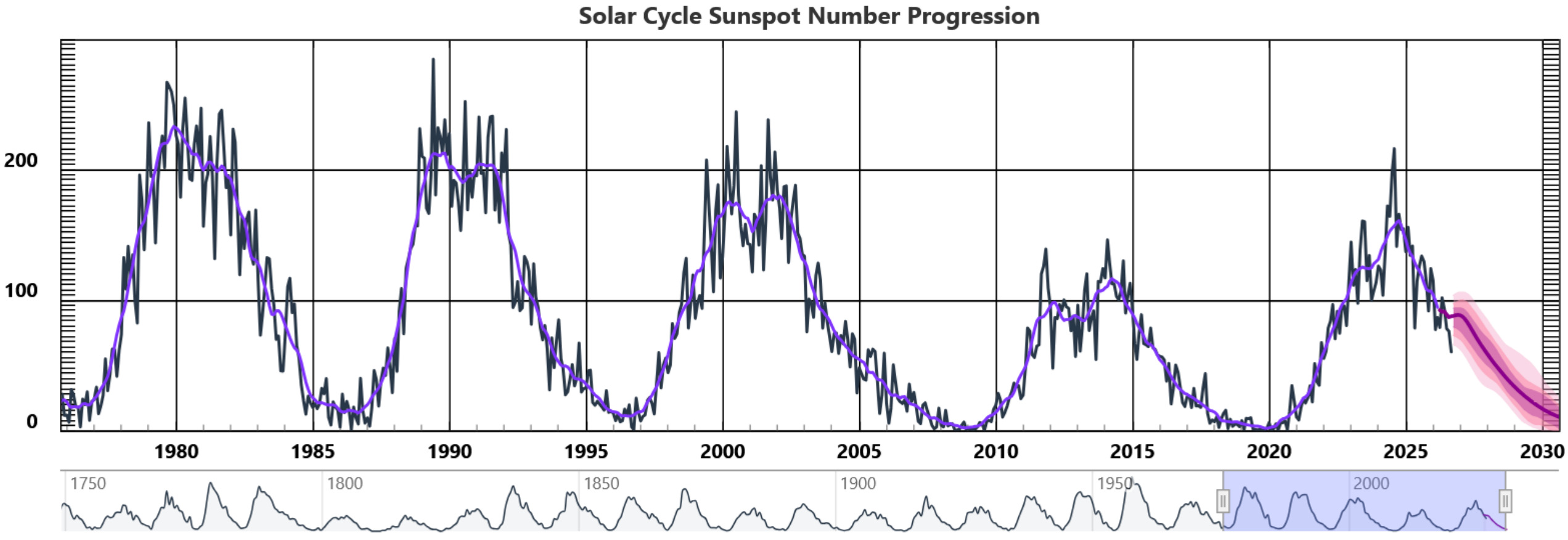




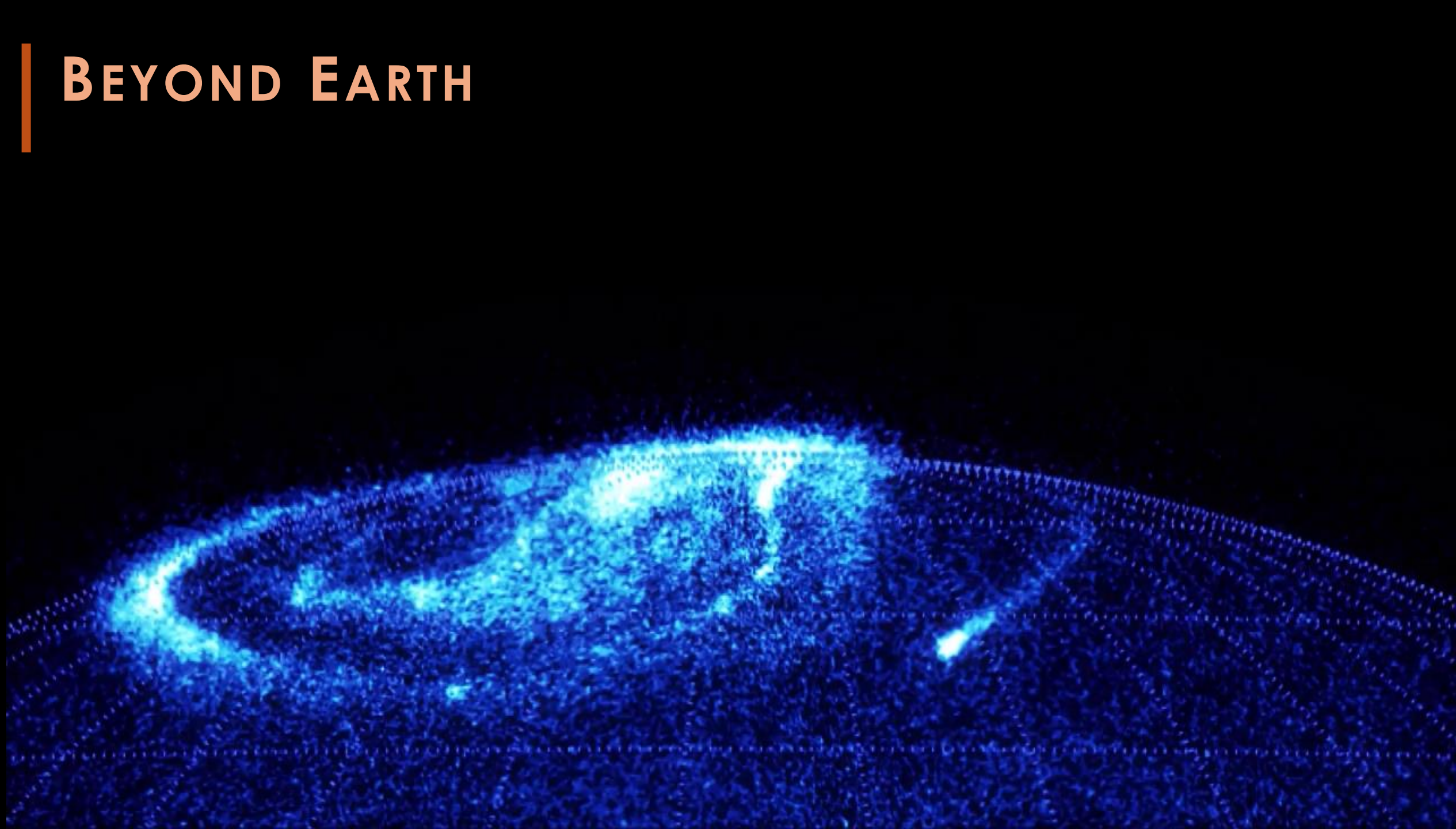
Brighton – October 2026



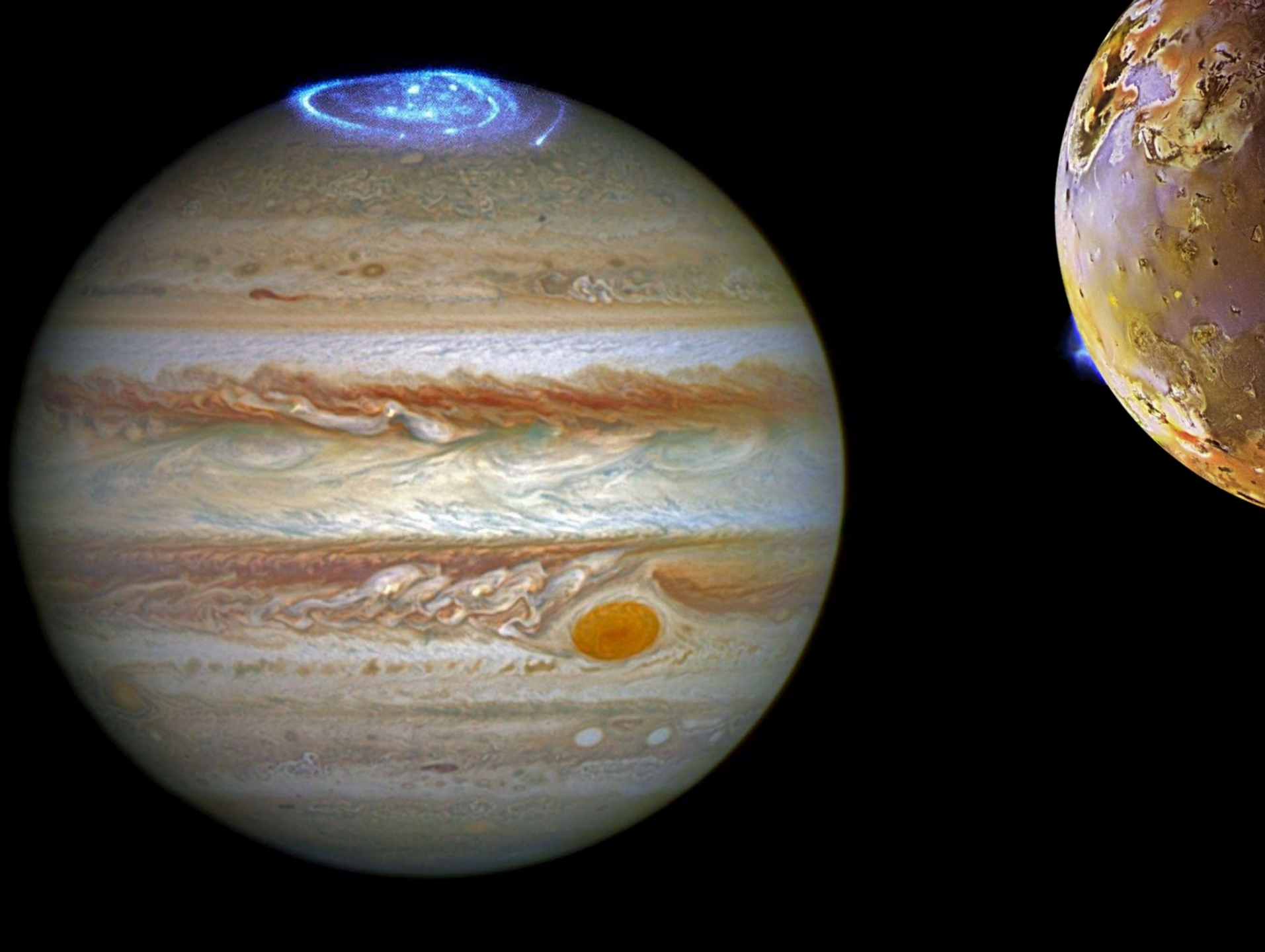
Solar Cycle


Solar Cycle Sunspot Number Progression
200
100
0
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
2030
1750
1800
1850
1900
1950
2000

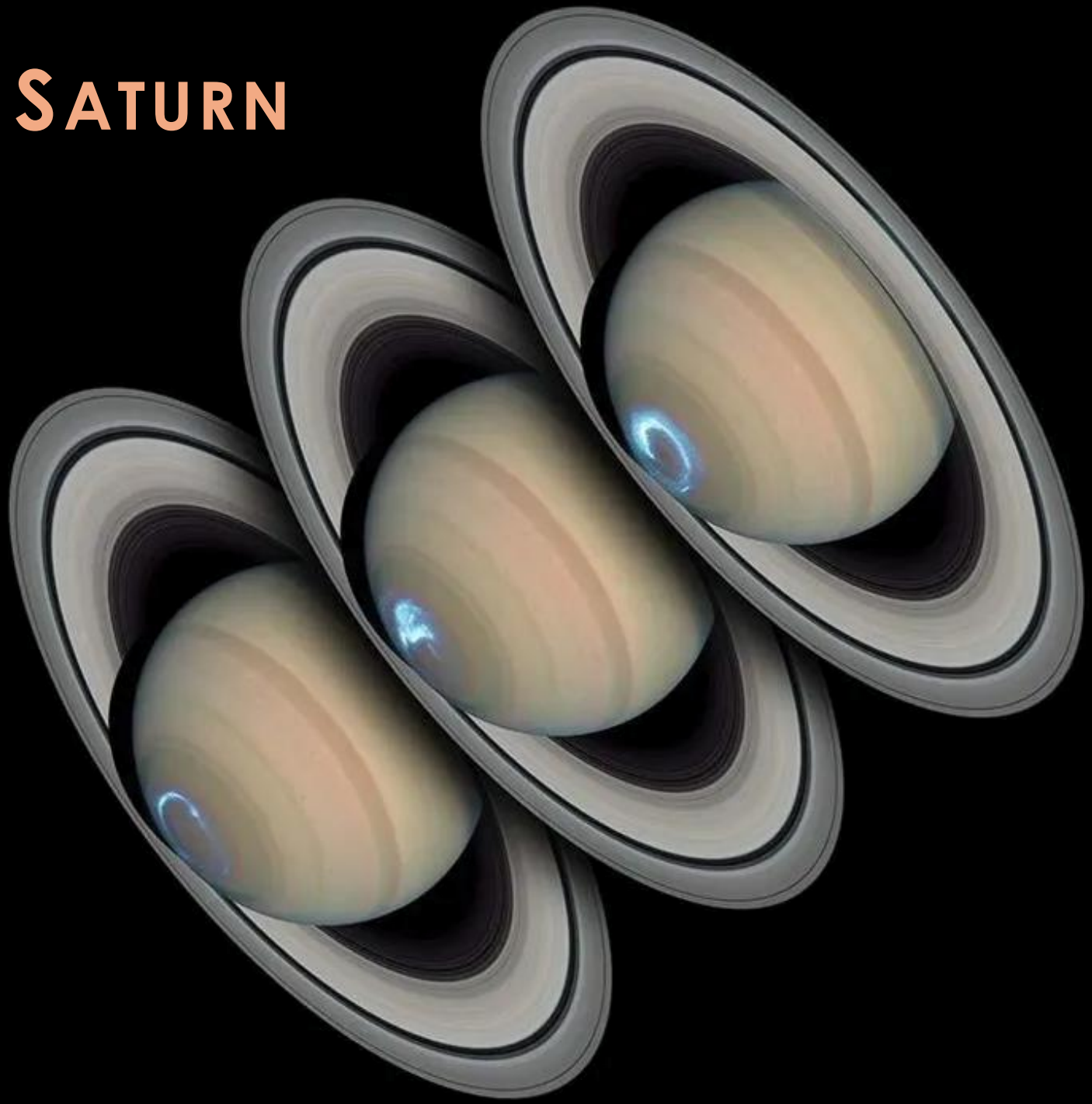
Beyond Earth



Jupiter



SATURN



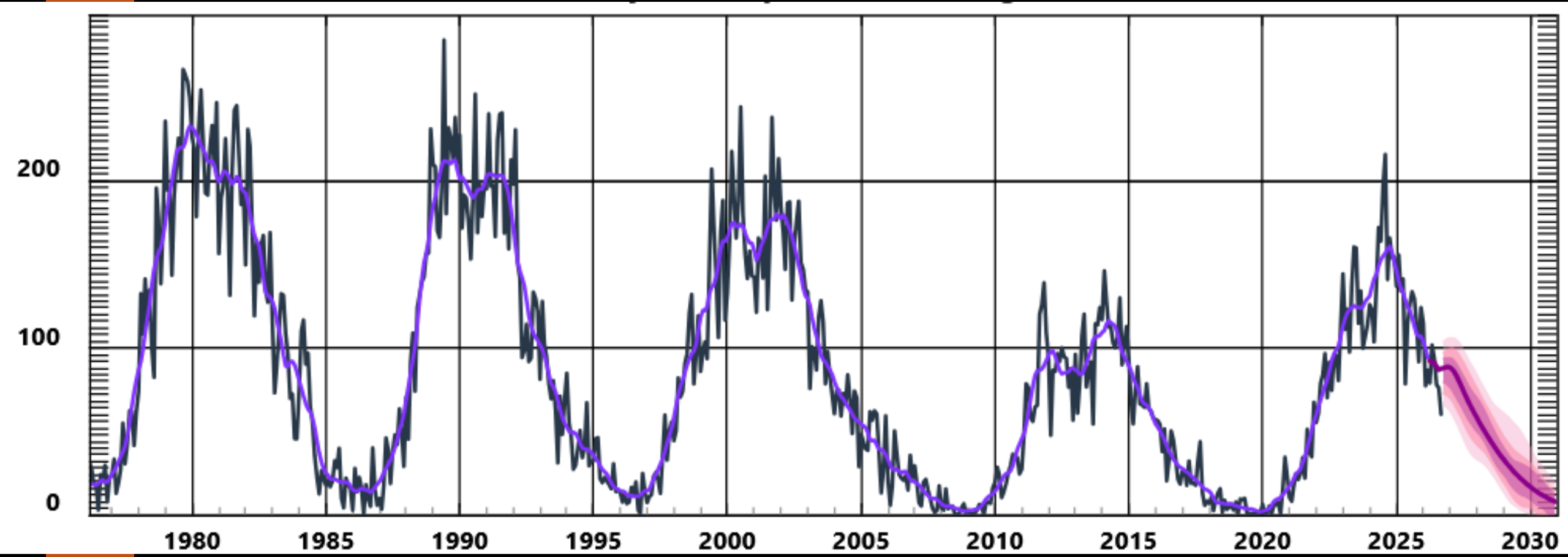
Exoplanets *have even been discovered through the* low-frequency radio-waves ($\lesssim$ 150 MHz) emitted by their aurora.

YZ Ceti
<12ly, 0.7 $M_\oplus$, 0.9$R_\oplus$, 0.02AU [2017]

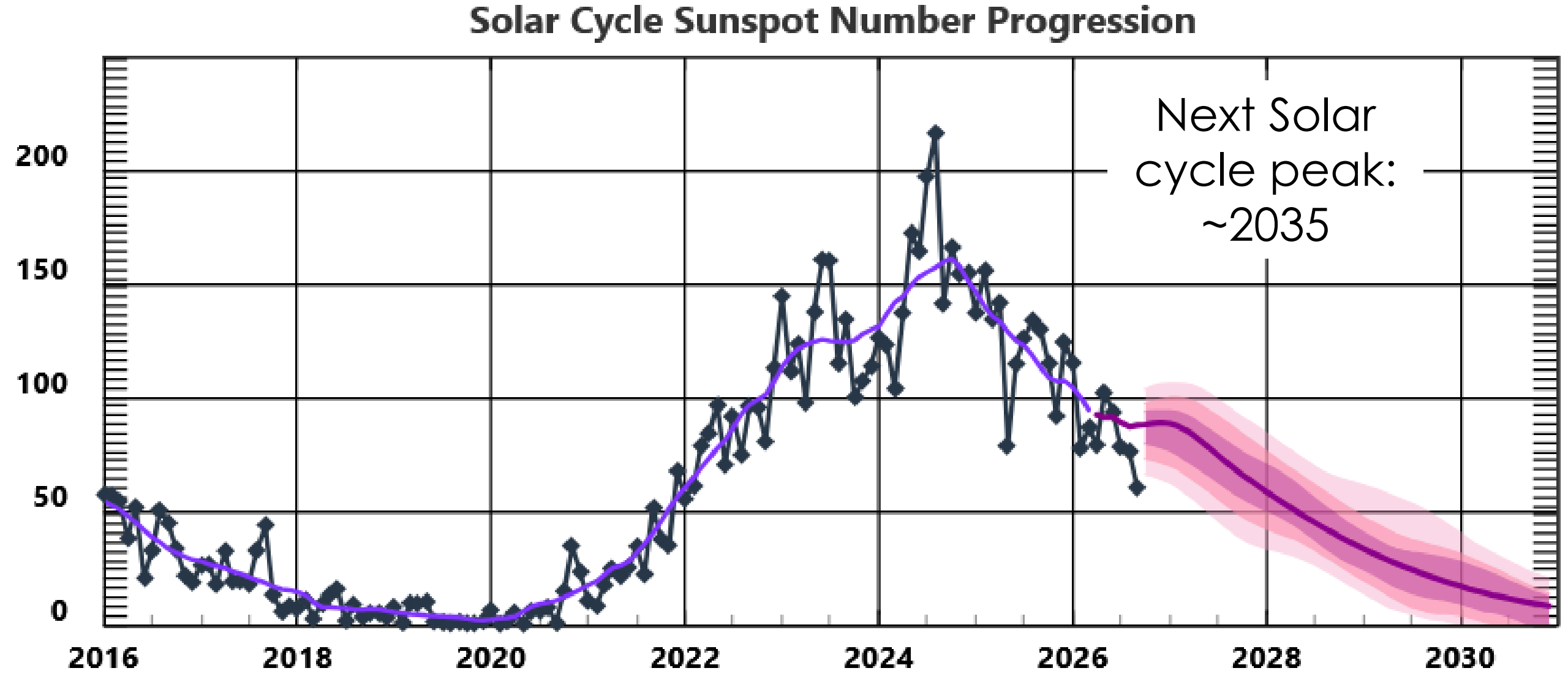
SIMP J0136+0933
20ly, 13 $M_\oplus$, 14 $R_\oplus$ [2003]

GJ 1151
26ly, $\leq 5M_\oplus$, 0.91$R_\oplus$ [2023]

Looking ahead to 2026/7



Looking ahead to 2026/7



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